

SamPave Engineering Suite

V2.0 Professional User & Engineering Guide

Version: v2.0-professional

Software Type: Pavement Engineering Decision Support Suite

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Important Notice: SamPave is a professional decision-support engineering tool. Final pavement design approval and field execution require review and sign-off by a qualified pavement engineer.

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SamPave V2.0 Quick Start Guide

CRITICAL SAFETY DISCLAIMER

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1. Installation and Launch (Fresh Machine)

To run SamPave, configure your local Python environment:

1. **Run Clean Setup:** Double-click [Setup.bat](#) in the project root. This script:

- Locates or user-installs Python 3.11+.
- Sets up a project-local virtual environment at `.venv`.
- Installs dependencies (PySide6, docx, openpyxl, sqlalchemy).
- Optionally builds the standalone executable `distSamPaveSamPave.exe` via PyInstaller.

2. **Launch the Suite:**

- Double-click [Launch.bat](#) to run from source code.
- Or execute the standalone binary at `distSamPaveSamPave.exe`.

2. 5-Minute Project Design Guide

Follow these steps to complete a flexible pavement design:

Step 1: Start a Project

1. Click **Create New Project** on the Project Hub.
2. Enter Project Name (e.g. **National Highway NH-580**), Client (**NHAI**), and Submitted By (**QA Engineer**).
3. Select Mix Type **DBM-II** and Binder Grade **VG-30**. Click **Create Project**.

Step 2: Input Traffic Data

1. Go to the **Traffic Analysis** tab.
2. Input:
 - CVPD = **1500**

- Growth Rate = 6.0%
- Design Life = 15 years
- VDF = 4.0
- LDF = 0.75

3. Click **Compute Traffic**. The cumulative traffic design (e.g. 38.29 MSA) will calculate instantly. Click **Save & Continue**.

Step 3: Input Subgrade Properties

1. Go to the **Subgrade Properties** tab.
2. Input laboratory soaked CBR = 4.0%.
3. Click **Calculate Modulus**. The subgrade resilient modulus (e.g. 40.0 MPa) will calculate. Click **Save**.

Step 4: Perform Catalogue Design

1. Go to the **Catalogue Design** tab.
2. Click **Lookup Catalogue Composition**.
3. The system queries the digitized database and populates the suggested thicknesses:
 - Bituminous Concrete (BC) = 40 mm
 - Dense Bituminous Macadam (DBM) = 160 mm
 - Wet Mix Macadam (WMM) = 250 mm
 - Granular Sub-base (GSB) = 300 mm
4. Click **Adopt Suggestion** to load these into the Structural Design panel.

Step 5: Verify via IITPAVE & Intelligence Checker

1. Go to the **Structural Pavement Design** tab.
2. Review the thickness profile.
3. Review the **Engineering Screening Score** and warnings on the screen (powered by the Intelligence Checker).
4. If a local copy of IITPAVE.exe is configured, click **Run Mechanistic Validation** to compute strains.

Step 6: Export DPR & Submission ZIP

1. Navigate to the **Submission Center**.
2. Change status to **Reviewed**.
3. Click **Lock Project** to lock the design parameters.
4. Click **Export DPR** to save the formatted Word report.
5. Click **Export Package (.zip)** to compile the delivery ZIP package.

SamPave V2.0 Professional Baseline — User Manual

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1. Introduction & System Overview

SamPave Engineering Suite V2.0 Professional Baseline is a Qt-based desktop application designed to streamline the flexible pavement design workflow in accordance with the Indian Roads Congress (IRC) standards, specifically **IRC:37-2018** (Guidelines for the Design of Flexible Pavements) and Marshall mix design standards (**IRC:111-2009**).

The application integrates traffic analysis, subgrade estimation, catalog lookup, layer structure customization, stabilized cement-treated base/sub-base design, engineering compatibility checker, and the IITPAVE mechanistic analysis engine into a unified database-driven desktop suite.

2. Module Directory

2.1 Project Hub & Project Creation

- **Purpose:** The central workspace manager. Allows creation, duplication, revisioning, deletion, and management of highway design projects.
- **Required Inputs:**
 - Name of Work (Project Name)
 - Client Name (e.g., NHAI, State PWD)
 - Consultant Name
 - Submitted By (Lead Engineer)
 - Agency/Contractor
 - Mix Type Specification (e.g., DBM-I, DBM-II, BC-I, BC-II)
 - Binder Grade Selection (VG-30, VG-40, PMB-70)
- **Engineering Practice Source:** Comes from client tender files, contract documents, work orders, and consulting agreements.

- **Logic & Design Rules:**

- Project records are persisted in a local SQLite database (`pavement.db`).
- Upon project locking, the state is frozen (`locked = 1`), preventing accidental edits to any calculation module.
- Revisioning clones child records and increments the revision number, linking back to the parent project ID.

- **Step-by-Step Usage:**

1. Launch SamPave and open the Project Hub.
2. Click **Create New Project**.
3. Enter project metadata (Work Name, Client, Submitted By).
4. Select the desired Mix Type and Binder Grade from the dropdowns.
5. Click **Create Project**. The app opens the Project Dashboard.

- **Output Meaning:** Establishes a database record with a unique Project ID that binds all subsequent traffic, structural, stabilized, and mix design data.

- **Common Mistakes:**

- Entering duplicate project names, causing confusion during audit.
 - Failing to set the correct binder grade or mix type early, which affects default elastic modulus parameters.
 - **Validation Method:** Confirm the project appears in the Hub list and database `projects` table has correct metadata.
 - **Safety Disclaimer:** Project metadata is for reporting trace purposes only. Ensure project boundaries and client titles match contract records exactly.
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2.2 Traffic Analysis & MSA Calculation

- **Purpose:** Computes the cumulative design traffic in Million Standard Axles (MSA) over the design life of the pavement.

- **Required Inputs:**

- Initial Commercial Vehicles Per Day (CVPD) in both directions.
- Annual Growth Rate (\$r\$ in %).
- Design Life (\$n\$ in years).
- Vehicle Damage Factor (VDF).
- Lane Distribution Factor (LDF).

- **Engineering Practice Source:**

- CVPD comes from a 7-day, 24-hour classified traffic volume count survey (IRC:9).
- Growth rate comes from econometric data or historic traffic trends.
- VDF comes from axle-load survey statistics or default tables in IRC:37.

- LDF comes from lane count configurations defined in IRC:37 Clause 4.4.

- **Formula & Logic:**

$$N = \frac{365 * A * [(1+r)^n - 1]}{r} * D * F * 10^{-6}$$

Where:

- N = Cumulative design traffic (MSA)
- A = Initial CVPD after construction $= P * (1+r)^x$ (where P is traffic count and x is years between count and completion)
- r = Annual growth rate decimal
- n = Design life (years)
- D = Lane Distribution Factor (LDF)
- F = Vehicle Damage Factor (VDF)
- *Linear Guard:* If $r = 0$, the formula simplifies to:

$$N = 365 * A * n * D * F * 10^{-6}$$

- **Step-by-Step Usage:**

1. Navigate to the **Traffic Analysis** page.
2. Input the CVPD, annual growth rate, design life, VDF, and LDF.
3. Click **Compute Traffic**. The page displays the cumulative design traffic (MSA) and growth factor.
4. Click **Save & Continue**.

- **Output Meaning:** The computed design traffic (MSA) determines the structural catalog thickness requirements.

- **Common Mistakes:**

- Entering the growth rate as a decimal (e.g., entering 0.05 instead of 5.0 for 5%).
- Confusing LDF values (e.g., using 0.75 for a single-carriageway 2-lane road instead of 0.50).

- **Validation Method:** Hand-calculate using Eq. above and compare. Error must be 0.0000%.

- **Safety Disclaimer:** Traffic forecasts are subject to high variance. Sensitivity analysis for VDF and growth rate should be conducted.
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2.3 CBR & Subgrade Mr Calculation

- **Purpose:** Translates laboratory soaked California Bearing Ratio (CBR) values into the design resilient modulus (M_r) of the subgrade.

- **Required Inputs:**

- Soaked CBR (%) of the subgrade soil.

- **Engineering Practice Source:** Laboratory 4-day soaked CBR tests conducted on representative subgrade soil samples (IS 2720 Part 16).

- **Formula & Logic:**

- For $\text{CBR} \leq 5.0\%$:

$$M_r = 10 * \text{CBR} \text{ (in MPa)}$$

- For $\text{CBR} > 5.0\%$:

$$M_r = 17.6 * \text{CBR}^{0.64} \text{ (in MPa)}$$

- *Lower boundary guard:* CBR must be $> 0.0\%$. Values $\leq 0.0\%$ raise a **ValueError**.

- **Step-by-Step Usage:**

1. Navigate to the **Subgrade Properties** page.
2. Enter the design subgrade soaked CBR (%).
3. Click **Calculate Modulus**. The page displays the resilient modulus (M_r) in MPa.
4. Click **Save**.

- **Output Meaning:** Represents the elastic support capability (M_r) of the subgrade soil, which is the foundational support layer in the pavement system.

- **Common Mistakes:**

- Using unsoaked CBR values, which overestimate subgrade support under monsoon conditions.
- Entering soil classifications without laboratory CBR verification.

- **Validation Method:** Compare calculated M_r against empirical formulas.

- **Safety Disclaimer:** Subgrade modulus is highly moisture-sensitive. Ensure appropriate drainage design accompanies the selected modulus value.

2.4 IRC Catalogue Design

- **Purpose:** Automatically suggests a structural pavement composition based on the computed design traffic (MSA) and subgrade CBR.

- **Required Inputs:** None (automatically loaded from Traffic and Subgrade modules).

- **Engineering Practice Source:** The database represents digitized skeleton plates of **IRC:37-2018 Plates 1 to 4**.

- **Logic & Design Rules:**

- The catalog suggestions scale thicknesses logically: higher traffic and weaker subgrade soils trigger thicker pavement profiles.
- Granular layers (WMM and GSB) provide the sub-base/base support.
- Dense Bituminous Macadam (DBM) and Bituminous Concrete (BC) provide the bituminous surface.
- *Capping Warning:* Design traffic exceeding 150 MSA generates an engineering warning that the catalog limits are exceeded, and mechanistic validation is mandatory.

- **Step-by-Step Usage:**

1. Navigate to the **Catalogue Design** tab.
 2. The page reads the active design MSA and subgrade CBR.
 3. Click **Lookup Catalogue Composition**.
 4. The system displays the recommended thicknesses for BC, DBM, WMM, and GSB.
- **Output Meaning:** Provides the standard baseline thickness composition suggested by the IRC code plates.
 - **Common Mistakes:**
 - Adopting catalog suggestions directly for construction without running a mechanistic strain verification.
 - **Validation Method:** Cross-check suggestions against the tables in IRC:37.
 - **Safety Disclaimer:** The catalog provides indicative designs. Regional variations, local aggregate quality, and climate factors must be evaluated.
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2.5 Structural Pavement Design

- **Purpose:** Allows customization of the pavement layer thicknesses, material properties (moduli), and Poisson's ratios, and performs structural calculations.
- **Required Inputs:**
 - Number of layers (BC, DBM, WMM, GSB, Subgrade).
 - Thickness of each layer (mm).
 - Elastic Modulus of each layer (MPa).
 - Poisson's ratio of each layer.
- **Engineering Practice Source:** Moduli and Poisson's ratios are sourced from laboratory dynamic modulus tests, resonant column tests, or default values in IRC:37 Clause 6.2.
- **Logic & Design Rules:**
 - Allows manual overrides of catalogue values to adjust to site-specific conditions.
 - Validates physical thickness limits (e.g. thickness must be ≥ 0).
- **Step-by-Step Usage:**
 1. Navigate to the **Structural Pavement Design** page.
 2. Review the suggested catalogue thickness composition.
 3. To change a layer thickness or modulus, edit the value directly in the grid.
 4. Click **Compute Structural Design**.
- **Output Meaning:** The customized layer profile, which serves as the geometry and elastic input for the IITPAVE mechanistic simulation.
- **Common Mistakes:**
 - Inputting a lower layer with a significantly higher modulus than the upper layer without crack-relief design (stiffness inversion risk).
 - Entering unrealistic Poisson's ratios (e.g. 0.50, which causes mathematical singularities).

- **Validation Method:** Confirm the total thickness is the sum of individual layer thicknesses.
- **Safety Disclaimer:** Custom elastic moduli must represent actual pavement materials. Stiffening a layer on paper does not translate to field performance without strict quality control.

2.6 Bituminous & Marshall Mix Design

- **Purpose:** Analyzes laboratory Marshall compaction samples to determine volumetric properties (\$V_a\$, \$VMA\$, \$VFB\$) and the Optimum Binder Content (OBC).
- **Required Inputs:**
 - Specific gravity of coarse aggregates, fine aggregates, and bitumen.
 - Blend ratios of aggregates.
 - Bulk specific gravity (\$G_{mb}\$) of compacted mix specimens at varying bitumen contents (\$P_b\$).
 - Maximum theoretical specific gravity (\$G_{mm}\$) of loose specimens.
 - Stability and flow readings.
- **Engineering Practice Source:** Laboratory Marshall mix design tests (ASTM D6927 / IRC:111).
- **Formula & Logic:**
 - Air Voids (\$V_a\$):

$$V_a = \left(1 - \frac{G_{mb}}{G_{mm}}\right) \times 100$$

- Voids in Mineral Aggregate (\$VMA\$):

$$VMA = 100 - \frac{G_{mb}}{P_s} \times G_{sb}$$

(where \$P_s = 100 - P_b\$ is the aggregate percentage, and \$G_{sb}\$ is the combined aggregate bulk specific gravity)

- Voids Filled with Bitumen (\$VFB\$):

$$VFB = \frac{VMA - V_a}{VMA} \times 100$$

- **OBC Determination:** Cubic spline interpolation calculates the OBC, representing the binder content that satisfies the target air voids (typically 4.0% for DBM/BC).

- **Step-by-Step Usage:**

1. Navigate to the **Mix Design** module.
2. Input laboratory aggregate specific gravities and blend ratios.
3. Enter aggregate weight data, loose specimen flask weight details (for \$G_{mm}\$), and bulk specimen measurements (for \$G_{mb}\$).
4. Input stability (kN) and flow (mm) measurements.
5. Click **Compute Mix Design**. The system generates volumetric plots and highlights the OBC.

- **Output Meaning:** The mix recipe (OBC) and volumetric checks confirming compliance with IRC:111/MORTH specifications.
 - **Common Mistakes:**
 - Inputting chaotic specific gravities, causing cubic splines to warp or yielding unphysical volumetric parameters.
 - **Validation Method:** Check calculated values against equations. Error must be 0.0000%.
 - **Safety Disclaimer:** Lab mix design parameters must be verified through trial tracks and field density testing.
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2.7 CTB/CTS Stabilized Pavement Wizard

- **Purpose:** Suggests and validates cement-treated base (CTB) and cement-treated sub-base (CTS) layers, evaluating thickness savings and cracking risks.
- **Required Inputs:**
 - Cement-Treated Base thickness (mm) and modulus (MPa).
 - Unconfined Compressive Strength (UCS in MPa) of the CTB.
 - Cement-Treated Sub-base thickness (mm) and modulus (MPa).
 - Granular Sub-base (GSB) support layer thickness (mm).
- **Engineering Practice Source:** Sourced from laboratory cement-aggregate mix trials (UCS tests) and specifications in IRC:37 Section 9.
- **Logic & Design Rules:**
 - Compares the CTB/CTS stack against conventional flexible structures.
 - Computes nominal thickness savings (typically 30-50% reduction in bituminous cover thickness).
 - Flags a warning if UCS is zero (missing UCS).
 - Flags modular ratio and thin stabilized layer risks.
- **Step-by-Step Usage:**
 1. Navigate to the **Stabilized Design** tab.
 2. Configure CTB thickness, modulus, and UCS.
 3. Configure CTS thickness and modulus.
 4. Click **Calculate Stabilized Composition**.
 5. Review side-by-side comparison.
- **Output Meaning:** Provides structural suggestions and thickness reductions for stabilized pavements.
- **Common Mistakes:**
 - Using high CTB stiffness directly over a weak subgrade support, causing fatigue cracking of the stabilized slab.
- **Validation Method:** Compare composition thickness totals against design criteria.

- **Safety Disclaimer:** Cement-stabilized bases are highly prone to shrinkage cracking. Crack-relief interlayers (e.g. SAMI or aggregate layer) are critical.

2.8 Engineering Intelligence Checker

- **Purpose:** Screens pavement designs for structural layer compatibility, modular ratios, and stiffness anomalies before report finalization.
- **Required Inputs:** None (automatically evaluates active composition).
- **Engineering Practice Source:** Sourced from pavement mechanics principles and IRC:37 Section 6.
- **Logic & Design Rules:**
 - **Engineering Screening Score:** Starts at 100.0, and deducts points for design risks:
 - *Unusual thickness:* BC outside 25-60mm (-5), DBM outside 50-200mm (-5), CTB/CTS ≤ 100 mm (-15).
 - *Non-standard Poisson's ratio:* Stabilized layer Poisson outside 0.15-0.35 (-10).
 - *Bad modular ratio:* Base-to-subbase modulus ratio ≥ 4.0 (-15).
 - *Stiff layer over weak support:* CTB/CTS over support ≤ 500 MPa (-20).
 - *Stiffness inversion:* Lower layer stiffer than upper layer (-15).
 - *Bottom layer over subgrade:* Modulus ratio ≥ 4.0 (-15).
- **Step-by-Step Usage:**
 1. Click **Run Intelligence Check** or review the results on the **Dashboard**.
 2. The Checker displays the **Engineering Screening Score (0-100)** and flags compatibility warnings.
- **Output Meaning:** An indicator of the design's structural harmony. High score means low risk.
- **Common Mistakes:**
 - Treating the score as a final approval or pass/fail stamp. It is a screening tool.
- **Validation Method:** Trace deductions using the rules above.
- **Safety Disclaimer:** A 100/100 score does not guarantee structural performance. Mechanistic validation (IITPAVE) remains mandatory.

2.9 IITPAVE Integration Manager

- **Purpose:** Orchestrates, validates, and parses execution files of the Indian Roads Congress mechanistic design engine (**IITPAVE.exe**).
- **Required Inputs:**
 - **IITPAVE.exe** executable binary path.
 - Elastic properties of the layers (thickness, modulus, Poisson).
 - Wheel load configuration (default 20,000 N per wheel at 0.8 MPa contact pressure).
- **Engineering Practice Source:** The calculation kernel (**IITPAVE.exe**) is standard software distributed by the Indian Roads Congress.

- **Logic & Design Rules:**

- **Decision Support Mode:** Enabled if **IITPAVE.exe** is missing. IITPAVE run is blocked, and placeholder reports are generated.

- **Mechanistic Verified Mode:** Enabled only when a valid path to **IITPAVE.exe** is configured, execution completes successfully, and output parser successfully extracts strains.

- **Step-by-Step Usage:**

1. Navigate to the **IITPAVE Manager** page.
2. Configure the path to your local **IITPAVE.exe** binary.
3. Click **Validate Path**.
4. If valid, the system activates **Mechanistic Verified Mode** for all calculations.

- **Output Meaning:** Confirms execution compatibility and manages files (**iitp_inp.txt**, **iitp_out.txt**).

- **Common Mistakes:**

- Providing an invalid path or a corrupt file.
 - **Validation Method:** Check for version detection and path status on screen.
 - **Safety Disclaimer:** Always run on a licensed copy of **IITPAVE.exe**.
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2.10 DPR/Word Report Export

- **Purpose:** Generates a professional Detailed Project Report (DPR) containing all traffic, structural, stabilized, and mix design data.

- **Required Inputs:**

- Report ID
- Assumptions and limitations metadata
- Engineer sign-off checklist selections

- **Engineering Practice Source:** Sourced from standard DPR formats required by public works authorities.

- **Logic & Design Rules:**

- Exports directly to a formatted **.docx** file using native document generators.
- Forces inclusion of disclaimers, assumptions, limitations, and check lists.

- **Step-by-Step Usage:**

1. Click **Export DPR** on the project screen.
2. Review and select checklist items (e.g. "Traffic count verified").
3. Provide sign-off comments.
4. Choose the save path.
5. Click **Generate**.

- **Output Meaning:** A formatted Word document ready for internal review and consultant sign-off.

- **Common Mistakes:**

- Failing to review the generated document before submission.
 - **Validation Method:** Open the generated file and verify all tables are populated.
 - **Safety Disclaimer:** Generating a report does not imply professional approval. Sign-off spaces are manual and must be signed by hand.
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2.11 Submission Center & Project Locking

- **Purpose:** Locks project data to freeze design parameters and tracks consultancy review status.
 - **Required Inputs:**
 - Consultancy status selection (Draft, Under Review, Reviewed, Approved for Submission).
 - **Logic & Design Rules:**
 - Locking freezes calculations (**locked** = 1).
 - To edit a locked project, the user must click **Unlock** (requires confirming a warning) or create a new **Revision Copy** (incrementing revision number).
 - **Step-by-Step Usage:**
 1. Open the **Submission Center** tab.
 2. Select the desired **Review Status**.
 3. Click **Lock Project**.
 - **Output Meaning:** Restricts edits and secures calculation history.
 - **Common Mistakes:**
 - Unlocking a project and making silent edits without creating a new revision.
 - **Validation Method:** Try editing any input field on other pages; fields should be disabled.
 - **Safety Disclaimer:** "Approved for Submission" denotes internal consultancy workflow compliance only, not government or code certification.
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2.12 Submission ZIP Package

- **Purpose:** Packages the Word report, serialized JSON database inputs, raw IITPAVE files, and revision logs into a single package.
- **Required Inputs:** None.
- **Step-by-Step Usage:**
 1. Open the **Submission Center** page.
 2. Click **Export Package (.zip)**.
 3. Select the output directory.
 4. The system compiles the ZIP archive.

- **Output Meaning:** The delivery file containing both the human-readable report and machine-readable calculation history.
 - **Common Mistakes:**
 - Deleting the compiled Word report before archiving, causing a missing file error in the packager.
 - **Validation Method:** Extract the ZIP file and check for `README_Disclaimer.txt`, `ProjectInputs.json`, and the Word report.
 - **Safety Disclaimer:** The ZIP package is decision-support documentation. Final sign-off remains mandatory.
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3. Recommended Operational Settings

- **Dynamic modulus settings:** VG-30 bitumen binder dynamic modulus should typically range between 2000 and 3500 MPa at 35°C design temperature.
- **Granular layer ratios:** Modular ratio of base-to-subbase must be kept between 2.0 and 4.0 for structural compatibility.
- **Subgrade support:** Do not use CBR values below 2.0% without soil stabilization or subgrade borrow-soil replacement.

SamPave Engineering Training Guide — V2.0

Professional Baseline

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1. Pavement Design Philosophy (IRC:37-2018)

IRC:37-2018 guides the design of flexible pavements using a mechanistic-empirical (M-E) framework. Pavements are modeled as multi-layered elastic structures under a standard dual-wheel configuration. Design evaluation involves two core criteria:

1. **Bituminous layer fatigue cracking** (tensile strain at the bottom of the bituminous layer, ϵ_t).
2. **Subgrade rutting** (vertical compressive strain at the top of the subgrade, ϵ_v).

2. Mathematical Core & Structural Equations

2.1 Traffic Projections (Cumulative MSA)

Design traffic is calculated as cumulative Million Standard Axles (MSA) over the design life (N):

$$N = \frac{365 \cdot A \cdot [(1+r)^n - 1]}{r} \cdot D \cdot F \cdot 10^{-6}$$

Where:

- N = Cumulative design traffic (MSA)
- A = Initial commercial traffic (CVPD) after construction $= P \cdot (1+r)^x$ (where P is count traffic and x is years between count and construction completion)
- r = Annual traffic growth rate decimal
- n = Design life (years)
- D = Lane Distribution Factor (LDF)
- Single-lane roads: LDF = 1.0
- Two-lane single carriageway: LDF = 0.50 (each direction)

- Four-lane single carriageway: LDF = 0.40
- Dual carriageway 2-lane: LDF = 0.75
- Dual carriageway 3-lane: LDF = 0.60
- Dual carriageway 4-lane: LDF = 0.45
- VF = Vehicle Damage Factor (VDF), converting commercial vehicles to equivalent standard axles.
- *Linear Case*: When growth rate $r = 0\%$, traffic accumulates linearly:

$$N = 365 * A * n * D * F * 10^{-6}$$

2.2 Subgrade Support (Resilient Modulus, M_r)

Subgrade resilient modulus (M_r) represents soil stiffness under repeated wheel loads. Per **IRC:37-2018 Annex E**:

- For soaked CBR (%) $\leq 5.0\%$:

$$M_r = 10 * CBR \text{ (in MPa)}$$

- For soaked CBR (%) $> 5.0\%$:

$$M_r = 17.6 * CBR^{0.64} \text{ (in MPa)}$$

- *Soil Cap Check*: Soil with soaked CBR $< 2.0\%$ must be replaced or stabilized with lime/cement.

2.3 Mechanistic Verification Strain Criteria

Pavements are evaluated at critical strain points under dual-wheel load (20,000 N per wheel at 0.8 MPa contact pressure).

Fatigue Life of Bituminous Layers (N_f)

Fatigue life represents the number of standard axle repetitions allowed before 20% cracking of the pavement surface. Sourced from **IRC:37-2018 Eq. 3.1 & 3.2**:

$$N_f = 0.0571 * \beta_f * \left(\frac{1}{\epsilon_t} \right)^{3.89} * \left(\frac{1}{E} \right)^{0.854}$$

Where:

- ϵ_t = Tensile strain at the bottom of the bituminous layer.
- E = Elastic modulus of the bituminous mix (MPa).
- β_f = Fatigue calibration factor (depends on binder type and air voids).

Rutting Life of Subgrade Soil (N_r)

Rutting life represents standard axle repetitions allowed before 20mm rut depth accumulates:

$$N_r = 1.41 \times 10^{-8} \times \beta_r \times \left(\frac{1}{\epsilon_v} \right)^{4.533}$$

Where:

- ϵ_v = Vertical compressive strain at the top of the subgrade.
- β_r = Rutting calibration factor.

3. CTB/CTS Stabilized Design Principles

Stabilized pavements introduce a Cement-Treated Base (CTB) and a Cement-Treated Sub-base (CTS) layer to increase load-bearing capability. Sourced from **IRC:37-2018 Section 9**:

3.1 Layer Configuration & Modulus

- **Bituminous Cover:** Sized to protect the stabilized slab from traffic wear and moisture infiltration.
- **CTB Modulus:** Sourced from laboratory flexural strength testing, typical design range: 3000 to 10000 MPa.
- **CTS Modulus:** Design range: 1000 to 4000 MPa.
- **GSB Layer:** Placed under the CTS to provide drainage and construction platforms.

3.2 Cracking Risks

Cement-treated materials are highly brittle and susceptible to:

1. **Shrinkage Cracking:** Occurs during cement hydration. Requires strict moisture control.
2. **Reflection Cracking:** Cracks in the CTB slab migrate upwards into the bituminous cover. Avoided by using SAMIs (Stress Absorbing Membrane Interlayers) or a granular sandwich layer.
3. **Tensile Fatigue:** Bottom tensile strain in the CTB slab must not exceed flexural capacity.

4. Marshall Mix Design (Volumetrics)

Marshall mix design determines the Optimum Binder Content (OBC) by checking the volumetric properties of compacted asphalt mixes (per **MORTH Section 500 / IRC:111**).

4.1 Volumetric Equations

- **Air Voids (V_a):**

$$V_a = \left(1 - \frac{G_{mb}}{G_{mm}} \right) \times 100$$

Where G_{mb} is bulk specific gravity and G_{mm} is maximum theoretical specific gravity of the compacted mix.

- **Voids in Mineral Aggregate (\$VMA\$):**

$$VMA = 100 - \frac{G_{mb}}{G_{mm}} \times P_s$$

Where $P_s = 100 - P_b$ is the aggregate percentage and G_{sb} is the combined aggregate specific gravity.

- **Voids Filled with Bitumen (\$VFB\$):**

$$VFB = \frac{VMA - V_a}{VMA} \times 100$$

4.2 OBC Spline Fitting

SamPave utilizes a cubic spline interpolation model to fit volumetric parameters against binder content (P_b). The OBC is determined as the binder percentage where V_a is exactly equal to 4.0% (target air voids), provided the mix satisfies MORTH limits for stability (> 9 kN), flow (2-4 mm), and VFB (65-75%).

5. Engineering Screening Compatibility Rules

The **Engineering Intelligence Checker** monitors layer layouts against compatibility guidelines:

- **Base-to-Subbase Modular Ratio:**

$$E_{base} / E_{sub-base} \leq 4.0$$

- **Stiff Layer Support:** Stiff stabilized base ($E \geq 3000$ MPa) must not sit directly on granular support (< 500 MPa), to prevent slab cracking.
- **Stiffness Inversions:** Lower layer elastic modulus must not exceed $1.3 \times E_{upper}$, preventing unbonded plate action.
- **Poisson's Ratio:** Restricted to realistic material bounds (granular = 0.35, stabilized = 0.25, bituminous = 0.35).

SamPave Demo Workflow Guide — NH-99 Project

CRITICAL SAFETY DISCLAIMER

IMPORTANT:

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1. Project Background

This demo workflow walks through the pavement design of the **National Highway NH-99 Corridor** (4-lane divided carriageway, subgrade soaked CBR of 4.0%). The goal is to design a structurally sound flexible pavement with stabilized layers to optimize material usage and minimize bituminous cover thickness.

2. Step-by-Step Walkthrough

2.1 Project Creation

1. Open the Project Hub and click **Create New Project**.
2. Set:
 - **Work Name:** NH-99 4-Lane Divided Carriageway
 - **Client:** NHA1 Regional Office
 - **Consultant Name:** Pavement Consultants Group
 - **Submitted By:** QA Lead Auditor
 - **Mix Type:** DBM-II
 - **Binder Grade:** VG-30
3. Click **Create Project**.

2.2 Traffic count and MSA Calculation

1. Go to the **Traffic Analysis** tab.
2. Input raw traffic survey data:
 - **Initial Commercial Traffic (CVPD):** 2500 (derived from 7-day volume counts)
 - **Annual Traffic Growth Rate (%):** 6.5

- **Design Life (years):** 15
- **Vehicle Damage Factor (VDF):** 4.5 (from axle-load survey)
- **Lane Distribution Factor (LDF):** 0.75 (typical for 4-lane divided carriageways, per IRC:37 cl. 4.4)

3. Click **Compute Traffic**.

• *Result:* Cumulative traffic calculated = **66.25 MSA**.

• *Growth Factor:* **23.716**.

4. Click **Save & Continue**.

2.3 Subgrade CBR and Modulus calculation

1. Go to the **Subgrade Properties** tab.

2. Input laboratory soaked CBR values:

• **Subgrade Soaked CBR (%):** 4.0

3. Click **Calculate Modulus**.

• *Result:* Modulus (M_r) = **40.0 MPa** (calculated as $10 \times 4.0\%$, since $CBR \leq 5\%$).

4. Click **Save**.

2.4 Catalogue design lookup

1. Go to the **Catalogue Design** tab.

2. Click **Lookup Catalogue Composition**.

• *Result:* The catalog query finds the closest match for **CBR 3-5%, MSA 50-100**:

• Bituminous Concrete (BC): 40 mm

• Dense Bituminous Macadam (DBM): 170 mm

• Wet Mix Macadam (WMM): 250 mm

• Granular Sub-base (GSB): 300 mm

3. Click **Adopt Suggestion** to load this into the Structural design workspace.

2.5 Stabilized Design Wizard (CTB/CTS Option)

1. Go to the **Stabilized Design** tab.

2. Input stabilized layer target values:

• **Cement-Treated Base (CTB) Thickness:** 120 mm

• **CTB Modulus:** 5000 MPa

• **CTB 28-day UCS Strength:** 4.5 MPa

- **Cement-Treated Sub-base (CTS) Thickness:** 100 mm
- **CTS Modulus:** 3000 MPa
- **Granular Sub-base (GSB) Thickness:** 150 mm
- **Bituminous Cover Thickness:** 100 mm

3. Click **Calculate Stabilized Composition**.

- **Result:** Nominal bituminous thickness savings = **110 mm** (a 52% saving in expensive asphalt binders).
- **Comparison mode:** Decision Support Mode (unless **IITPAVE.exe** is configured).

2.6 Engineering Intelligence Screen

1. Navigate to the **Structural Pavement Design** tab.
2. Review the **Engineering Screening Score** and warnings on the right side:
 - **Stiff Layer over Weak Support:** CTB is placed directly over GSB. Since GSB modulus is 150 MPa (< 500 MPa), this triggers a warning: *"Very stiff stabilized layer 'CTB' is placed directly over weak support. Potential risk of cracking due to stiffness mismatch."*
 - **Stiffness Inversion:** Reviewed; layers follow structural tier order.
 - **Screening Score:** Displays **65/100** (Needs engineering review).
3. Review findings and note in the design diary that shrinkage interlayers will be specified.

2.7 IITPAVE Mechanistic Verification (If exe configured)

1. If **IITPAVE.exe** is configured in the **IITPAVE Manager**, click **Run Mechanistic Validation**.
2. The engine runs in the background and parses the strain outputs:
 - Fatigue tensile strain (ϵ_t) at bottom of bituminous layer.
 - Vertical compressive strain (ϵ_v) at top of subgrade.
3. The system confirms if fatigue and rutting lives exceed design requirements.

2.8 Review checklist and Project Locking

1. Navigate to the **Submission Center**.
2. Complete the **Consultant Checklist**:
 - Check: *Traffic counts verified*
 - Check: *Subgrade soil soaked CBR test verified*
 - Check: *Layer modular ratios reviewed*
3. Set review status to **Reviewed**.
4. Click **Lock Project** to secure the design parameters from modification.

2.9 DPR Export and Submission ZIP Package

1. Click **Export DPR**. Choose file name `NH_99_Detailed_Project_Report.docx` and click **Save**.
2. Click **Export Package (.zip)**. Choose file name `NH_99_Submission_Package.zip` and click **Save**.
3. The ZIP package compiles:
 - `NH_99_Detailed_Project_Report.docx` (Word DPR)
 - `ProjectInputs.json` (Database snapshot)
 - `README_Disclaimer.txt` (Disclaimer text)
 - `ApprovalSheet.txt` (Checks sheet)

SamPave Design Auditor & Checker Validation Checklist

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This checklist is designed for senior design auditors, peer checkers, and government examiners to verify the integrity of pavement designs processed using the SamPave Engineering Suite.

1. Input Verification Check

- [] **Classified Traffic Volume Counts:** Verify that initial CVPD is sourced from a 7-day volume count survey (per IRC:9) and has been adjusted for the years between count and construction completion.
- [] **Traffic Growth Rate:** Confirm the annual growth rate ($r\%$) matches local masterplans or econometric projections. Watch out for unrealistically high projections ($r > 8.0\%$).
- [] **Design Life (n):** Verify design life matches road category specifications (typically 15 years for National Highways/State Highways, 20 years for expressways).
- [] **Vehicle Damage Factor (VDF):** Cross-check VDF against regional axle-load survey studies. If default tables are used, confirm the traffic volume class matches MORTH guidelines.
- [] **Lane Distribution Factor (LDF):** Confirm LDF matches road configuration (e.g. 0.75 for 4-lane divided carriageway).
- [] **Soaked CBR:** Confirm the subgrade soil CBR value represents a laboratory 4-day soaked test (per IS:2720 Part 16) on the worst-performing borrow-pit soil samples.

2. Structural Layer Compatibility Check

- [] **Base-to-Subbase Modular Ratio:** Confirm that the elastic modulus ratio between WMM (granular base) and GSB (granular sub-base) is between 2.0 and 4.0.
- [] **Stiffness Inversion:** Verify that lower layers are not stiffer than upper layers (e.g., granular sub-base modulus must not exceed granular base modulus).
- *Exception:* Bituminous cover over a Cement-Treated Base (CTB) slab is not considered a stiffness inversion.

- [] **Stiff CTB Over Granular Layer:** If a Cement-Treated Base ($E \geq 5000$ MPa) is specified, confirm that the underlying sub-base is either a Cement-Treated Sub-base (CTS) or a high-strength GSB ($E \geq 500$ MPa). If placed directly on weak support ($E < 500$ MPa), confirm shrinkage-relief measures are detailed.
 - [] **Poisson's Ratio:** Confirm Poisson's ratios conform to typical elastic limits:
 - Bituminous Concrete / DBM: 0.35
 - Wet Mix Macadam / Granular Base: 0.35
 - Cement-Treated Base / Sub-base: 0.25
 - Subgrade Soil: 0.35
-

3. Mix Design Volumetrics & OBC Check

- [] **Air Voids (\$V_a\$):** Confirm air voids at the designated Optimum Binder Content (OBC) fall within the target design specifications (3.0% to 5.0%, typically 4.0% for DBM/BC).
 - [] **Void in Mineral Aggregate (\$VMA\$):** Verify \$VMA\$ complies with minimum requirements (typically $\geq 12.0\%$ for DBM with 37.5mm nominal size and $\geq 13.0\%$ for 26.5mm size, per MORTH Table 500-11).
 - [] **Void Filled with Bitumen (\$VFB\$):** Check that \$VFB\$ is within the specified range (65% to 75% for DBM/BC).
 - [] **Stability & Flow:** Verify stability is > 9.0 kN and flow is between 2.0 mm and 4.0 mm.
-

4. IITPAVE Mechanistic Verification

- [] **Production Mode:** Confirm the design has been processed in **Mechanistic Verified Mode** (via local `IITPAVE.exe` runs) rather than Decision Support fallback mode.
 - [] **Tensile Strain (\$epsilon_t\$):** Confirm the tensile strain at the bottom of the bituminous layer does not exceed the fatigue life design strain.
 - [] **Vertical Compressive Strain (\$epsilon_v\$):** Confirm the vertical strain at the top of the subgrade does not exceed the rutting life design strain.
-

5. Review Checklist and Project Sign-Off

- [] **Project Lock State:** Verify the project is **Locked** in the Submission Center to prevent further modifications.
- [] **Submission ZIP Package:** Verify the ZIP archive exists and contains all required output logs and calculations.
- [] **Independent Engineer Sign-off:** Ensure the final report bears the manual signature and seal of a licensed Professional Pavement Engineer.